

Write all the answers in the answer book.

1. Find the convolution between $f_1(t)$ and $g_1(t)$ where

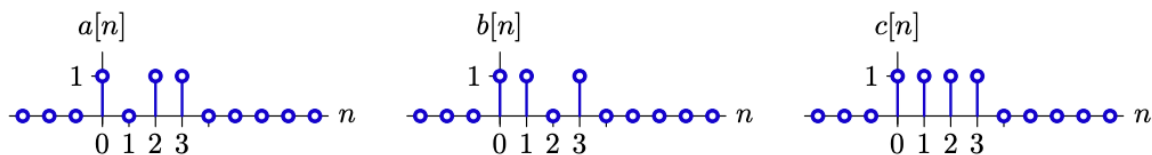
$$f_1(t) = \delta(t) + \delta(t-1) + \delta(t-2)$$

$$g_1(t) = \delta(t) + \delta(t-1) - \delta(t-2)$$

2. Determine the convolution of the two signals ($f_2(t)$ and $g_2(t)$) using graphical interpretation

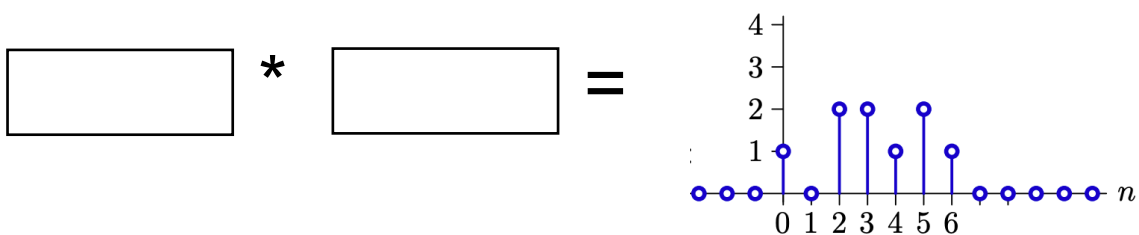
$$f_2(t) = \begin{cases} -\sin(\pi t); & |t| \leq 1 \\ 0; & |t| > 1 \end{cases}, \quad g_2(t) = \begin{cases} 1.25; & -0.5 \leq t \leq 1 \\ 0; & \text{Otherwise} \end{cases}$$

3. Given three signals: $a[n]$, $b[n]$ and $c[n]$ below.

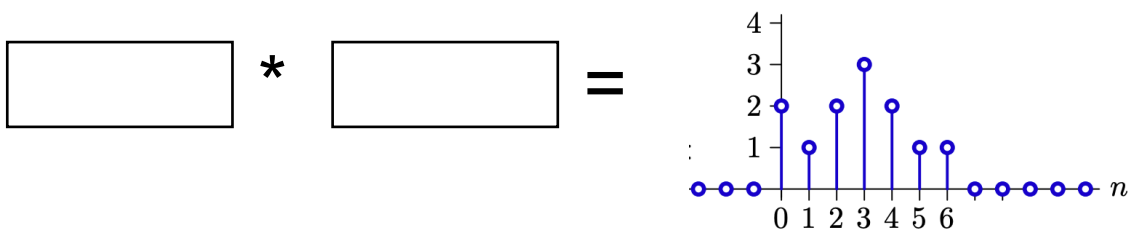


Fill in the input signals: $a[n]$, $b[n]$ or $c[n]$ or **X** (if none of them match the input from the convolution) in the blanket (**and in the answer book**). The results of each convolution pairs are displayed in the right hand side.

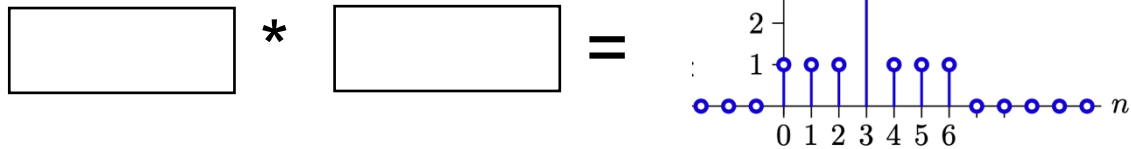
3.1)



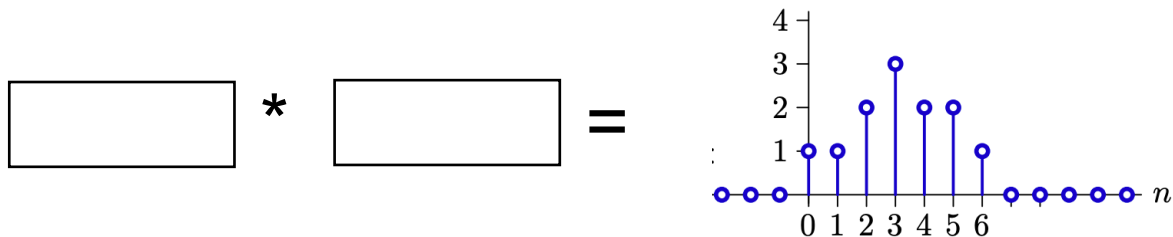
3.2)



3.3)



3.4)



4. Find the Fourier transform of the signal below

$$x(t) = te^{-|t|} \sin(2t)$$

5. Determine the convolution of the signals below

$$\text{sinc}(t) * \text{sinc}^2(t)$$

6. Prove the Parseval's theorem, that the energy in the Fourier Transform of a function is the same as the energy of the original function.

$$\int_{-\infty}^{\infty} |x(t)|^2 dt = \frac{1}{2\pi} \int_{-\infty}^{\infty} |X(\omega)|^2 d\omega$$

7. Consider the signal

$$x[n] = 1 + \sin\left(\frac{2\pi}{N}n\right) + 3\cos\left(\frac{2\pi}{N}n\right) + \cos\left(\frac{4\pi}{N}n + \frac{\pi}{2}\right)$$

Plot the real and imaginary part of the Fourier coefficients for $N = 5$

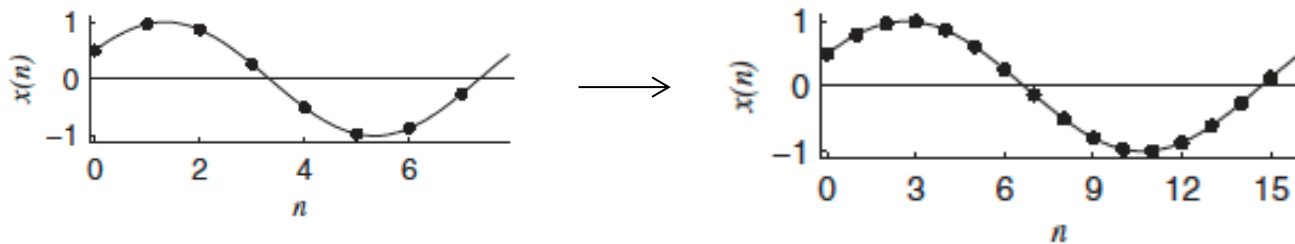
8. Prove that

$$\mathcal{DTFT}\{u[n]\} = \pi \sum_{k=-\infty}^{\infty} \delta(\omega - 2\pi k) + \frac{1}{1 - e^{j\omega}}$$

9. Find DFT of the signal below

$$x[n] = -2 + 3 \cos\left(\frac{2\pi}{4}n + \frac{\pi}{6}\right) - 2\cos(\pi n)$$

10. Explain the process of interpolation application of Fourier transform as shown in the figure below.



11. If we sampling signal $\sin(2\pi f t)$ with sampling rate f_s and $\frac{f_s}{2} < f < f_s$. Explain why we observe that is $-\sin(2\pi(f - f_s)t)$ after DAC converter